

Probability and Statistics

MATH-361

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Organizing Data

Suppose a researcher wished to do a study on the ages of the top 50 wealthiest people in the world. The researcher first would have to get the data on the ages of the people. In this case, these ages are listed in Forbes Magazine. When the data are in original form, they are called raw data and are listed next

49	57	38	73	81
74	59	76	65	69
54	56	69	68	78
65	85	49	69	61
48	81	68	37	43
78	82	43	64	67
52	56	81	77	79
85	40	85	59	80
60	71	57	61	69
61	83	90	87	74

Since little information can be obtained from looking at raw data, the researcher organizes the data into what is called a frequency distribution.

A **frequency distribution** is the organization of raw data in table form, using classes and frequencies.

Each raw data value is placed into a quantitative or qualitative category called a class. The frequency of a class then is the number of data values contained in a specific class. A frequency distribution is shown for the preceding data set

Class limits	Tally	Frequency
35-41	///	3
42-48	///	3
49-55	////	4
56-62	 	10
63-69	 	10
70-76		5
77-83	 	10
84-90		5
		Total 50

The classes in this distribution are 35-41, 42-48, etc. These values are called class limits. The data values 35, 36, 37, 38, 39, 40, 41 can be tallied in the first class; 42, 43, 44, 45, 46, 47, 48 in the second class; and so on.

Types of frequency distributions

Categorical frequency distribution The categorical frequency distribution is used for data that can be placed in specific categories, such as nominal- or ordinal-level data.

Grouped frequency distribution When the range of the data is large, the data must be grouped into classes that are more than one unit in width, in what is called a grouped frequency distribution.

Twenty-five army inductees were given a blood test to determine their blood type. The data set is

A, B, B, AB, O, O, O, B, AB, B, B, B,
O, A, O, A, O, O, O, AB, AB, A, O, B, A.

Construct a frequency distribution for the data.

A Class	B Tally	C Frequency	D Percent
A	////	5	20
B	//// //	7	28
O	//// ////	9	36
AB	////	4	16
		Total 25	100

A distribution of the number of hours that boat batteries lasted is the following

Class limits	Class boundaries	Tally	Frequency
24–30	23.5–30.5	///	3
31–37	30.5–37.5	/	1
38–44	37.5–44.5	////	5
45–51	44.5–51.5	//// ////	9
52–58	51.5–58.5	//// /	6
59–65	58.5–65.5	/	1
			25

Class limits	Class boundaries	Tally	Frequency
24–30	23.5–30.5	///	3
31–37	30.5–37.5	/	1
38–44	37.5–44.5	////	5
45–51	44.5–51.5	//// //	9
52–58	51.5–58.5	//// /	6
59–65	58.5–65.5	/	1
			25

The values 24 and 30 of the first class are called **class limits**

The lower class limit is 24; it represents the smallest data value that can be included in the class.

Class limits	Class boundaries	Tally	Frequency
24–30	23.5–30.5	///	3
31–37	30.5–37.5	/	1
38–44	37.5–44.5	///	5
45–51	44.5–51.5	/// ////	9
52–58	51.5–58.5	/// /	6
59–65	58.5–65.5	/	1
			25

The values 24 and 30 of the first class are called **class limits**

The lower class limit is 24; it represents the smallest data value that can be included in the class.

The upper class limit is 30; it represents the largest data value that can be included in the class.

Class limits	Class boundaries	Tally	Frequency
24–30	23.5–30.5	///	3
31–37	30.5–37.5	/	1
38–44	37.5–44.5	////	5
45–51	44.5–51.5	////	9
52–58	51.5–58.5	//// /	6
59–65	58.5–65.5	/	1
			25

The lower class limit is 24; it represents the smallest data value that can be included in the class.

The upper class limit is 30; it represents the largest data value that can be included in the class.

Class limits	Class boundaries	Tally	Frequency
24–30	23.5–30.5	///	3
31–37	30.5–37.5	/	1
38–44	37.5–44.5	////	5
45–51	44.5–51.5	//// ///	9
52–58	51.5–58.5	//// /	6
59–65	58.5–65.5	/	1
			25

The lower class limit is 24; it represents the smallest data value that can be included in the class.

The upper class limit is 30; it represents the largest data value that can be included in the class.

The numbers in the second column are called class boundaries.

These numbers are used to separate the classes so that there are no gaps in the frequency distribution. The gaps are due to the limits; for example, there is a gap between 30 and 31.

Boundary between 30 and 31 is given by

$$\frac{31 + 30}{2} = 30.5$$

The basic rule of thumb is that the class limits should have the same decimal place value as the data, but the class boundaries should have one additional place value and end in a 5.

Find the boundaries by subtracting 0.5 from 31 (the lower class limit) and adding 0.5 to 37 (the upper class limit).

$$\text{Lower limit} - 0.5 = 31 - 0.5 = 30.5 = \text{lower boundary}$$

$$\text{Upper limit} + 0.5 = 37 + 0.5 = 37.5 = \text{upper boundary}$$

Class width

The class width for a class in a frequency distribution is the difference between lower(or upper) class limits of consecutive classes

lower class limit of one class – lower class limit of the next class.

What is the class width in the preceding distribution on the duration of boat batteries ?

Range of Data

Range= Highest Value - Lowest Value

The following data represents the actual liquid weight in 16 "twelve-ounce" cans. Construct a frequency distribution with four classes from this data.

11.95	11.91	11.86	11.94	12.00	11.93	12.00	11.94
12.10	11.95	11.99	11.94	11.89	12.01	11.99	11.94

Step#1: Low score=11.86, high score=12.10

Step#2: Number of classes=4(given)

Step#3: The accuracy is two decimal places.

Step#4: Compute the lower and upper bounds.

$$\text{lower bound} = \frac{12.10 - 11.86}{4} = 0.06$$

$$\text{upper bound} = \frac{12.10 - 11.86}{4 - 1} = 0.08$$

We can

use any number bigger than 0.06, but not more than 0.08.

Step#5: Chose 0.08. By adding 0.08 to 11.86 repeatedly, we obtain the lower class limits: 11.86, 11.94, 12.02, 12.10.

Step#6: The first upper class limit is the largest number with the same accuracy as the data that is just below the second lower class limit. In this case, the number is 11.93. The other upper class limits are found by adding 0.08 repeatedly to 11.93, until there are 4 upper class limits.

The lower class limits: 11.86, 11.94, 12.02, 12.10.

The upper class limits: 11.93, 12.01, 12.09, 12.17.

Next, for each member of the data set, we decide which class contains it and then put a tally mark by that class. The numbers corresponding to these tallies gives us the class frequencies.

Class	Tally	Frequency
11.86-11.93		4
11.94-12.01		11
12.02-12.09		0
12.10-12.17		1

- ① There should be between 5 and 20 classes.
- ② It is preferable but not absolutely necessary that the class width be an odd number. This ensures that the midpoint of each class has the same place value as the data. The class midpoint X_m is obtained by adding the lower and upper boundaries and dividing by 2, or adding the lower and upper limits and dividing by 2:

$$X_m = \frac{\text{lower boundary} + \text{upper boundary}}{2}$$

or

$$X_m = \frac{\text{lower limit} + \text{upper limit}}{2}$$

- ③ The classes must be mutually exclusive.
- ④ The classes must be continuous.
- ⑤ The classes must be exhaustive. There should be enough classes to accommodate all the data.
- ⑥ The classes must be equal in width.

These data represent the record high temperatures in degrees Fahrenheit (°F) for each of the 50 states. Construct a grouped frequency distribution for the data using 7 classes.

112	100	127	120	134	118	105	110	109	112
110	118	117	116	118	122	114	114	105	109
107	112	114	115	118	117	118	122	106	110
116	108	110	121	113	120	119	111	104	111
120	113	120	117	105	110	118	112	114	114

Step-1 Determine the classes.

$$\text{class width} = \left\lceil \frac{\text{highest value} - \text{lowest value}}{\text{number of classes}} \right\rceil$$

Step-2 Tally the data.

Step-3 Find the numerical frequencies from the tallies. The completed frequency distribution is

Select a starting point for the lowest class limit. This can be the smallest data value or any convenient number less than the smallest data value. In this case, 100 is used. Add the width to the lowest score taken as the starting point to get the lower limit of the next class. Keep adding until there are 7 classes, as shown, 100, 105, 110, etc.

Subtract one unit from the lower limit of the second class to get the upper limit of the first class. Then add the width to each upper limit to get all the upper limits.

$$105 - 1 = 104$$

The first class is 100-104, the second class is 105-109, etc. Find the class boundaries by subtracting 0.5 from each lower class limit and adding 0.5 to each upper class limit:

Class limits	Class boundaries	Tally	Frequency
100–104	99.5–104.5	//	2
105–109	104.5–109.5	/// ///	8
110–114	109.5–114.5	/// /// /// ///	18
115–119	114.5–119.5	/// /// ///	13
120–124	119.5–124.5	/// //	7
125–129	124.5–129.5	/	1
130–134	129.5–134.5	/	1
			<u>1</u>
			$n = \Sigma f = 50$

Reading Class Limits, Boundaries and Midpoint

LOWER CLASS LIMITS

21, 31, 41, 51, 61, 71

UPPER CLASS LIMITS

30, 40, 50, 60, 70, 80

CLASS BOUNDARIES

* BETWEEN 30 & 31 = 30.5

* BETWEEN 40 & 41 = 40.5

* BETWEEN 50 & 51 = 50.5

ETC.

Frequency Distribution:
Ages of Academy Awards Best Actresses

Age of Actress	Frequency
21-30	28
31-40	30
41-50	12
51-60	2
61-70	2
71-80	2

CLASSES

MIDPOINTS

$(21+30)/2 = \underline{\underline{25.5}}$

$(31+40)/2 = \underline{\underline{35.5}}$

$(41+50)/2 = \underline{\underline{45.5}}$

ETC

CLASS WIDTH = 10

$31 - 21 = 10$

or

$40.5 - 30.5 = 10$

Sometimes it is necessary to use a cumulative frequency distribution. A cumulative frequency distribution is a distribution that shows the number of data values less than or equal to a specific value (usually an upper boundary). The values are found by adding the frequencies of the classes less than or equal to the upper class boundary of a specific class.

	Cumulative frequency
Less than 99.5	0
Less than 104.5	2
Less than 109.5	10
Less than 114.5	28
Less than 119.5	41
Less than 124.5	48
Less than 129.5	49
Less than 134.5	50

The data shown here represent the number of miles per gallon (mpg) that 30 selected four-wheel-drive sports utility vehicles obtained in city driving. Construct a frequency distribution, and analyze the distribution.

12	17	12	14	16	18
16	18	12	16	17	15
15	16	12	15	16	16
12	14	15	12	15	15
19	13	16	18	16	14

Procedure Table

Constructing a Grouped Frequency Distribution

Step 1 Determine the classes.

Find the highest and lowest values.

Find the range.

Select the number of classes desired.

Find the width by dividing the range by the number of classes and rounding up.

Select a starting point (usually the lowest value or any convenient number less than the lowest value); add the width to get the lower limits.

Find the upper class limits.

Find the boundaries.

Step 2 Tally the data.

Step 3 Find the numerical frequencies from the tallies, and find the cumulative frequencies.

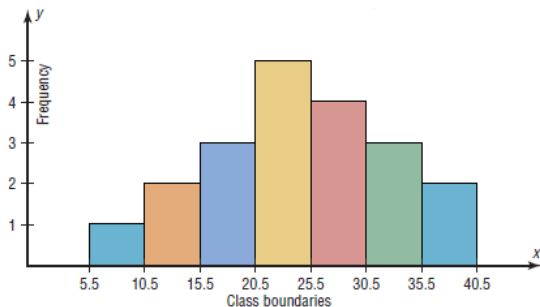
Make a grouped frequency distribution to the weights recorded to the nearest grams of 60 apples picket out at from a consignment.

106	107	76	82	109	107	115	93
187	95	123	125	111	92	86	70
126	68	130	129	139	119	115	128
100	186	84	99	113	204	111	141
136	123	90	115	98	110	78	185
162	178	140	152	173	146	158	194
148	90	107	181	131	75	184	104
110	80	118	82				

- ① Rang
- ② Class width
- ③ Number of classes
- ④ Lower and upper limits

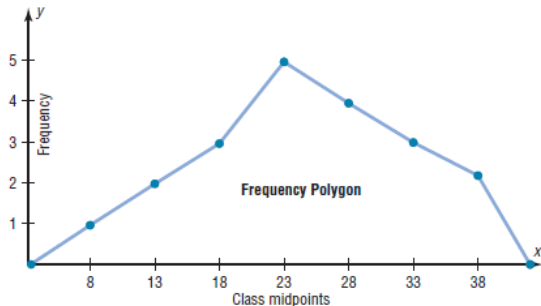
The **histogram** is a graph that displays the data by using contiguous vertical bars (unless the frequency of a class is 0) of various heights to represent the frequencies of the classes.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	20



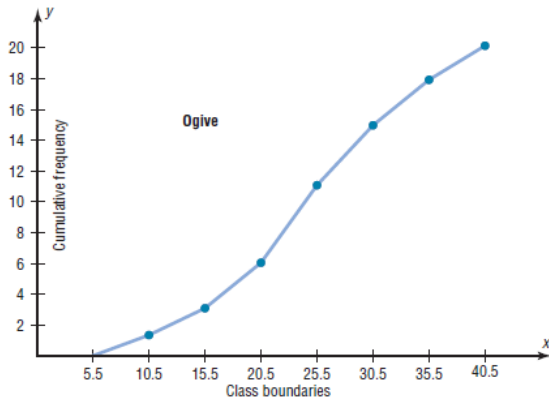
The **frequency polygon** is a graph that displays the data by using lines that connect points plotted for the frequencies at the midpoints of the classes. The frequencies are represented by the heights of the points.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	<u>20</u>



The **ogive** is a graph that represents the cumulative frequencies for the classes in a frequency distribution.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	<u>20</u>



Procedure Table

Constructing Statistical Graphs

- | | |
|---------------|--|
| Step 1 | Draw and label the x and y axes. |
| Step 2 | Choose a suitable scale for the frequencies or cumulative frequencies, and label it on the y axis. |
| Step 3 | Represent the class boundaries for the histogram or ogive, or the midpoint for the frequency polygon, on the x axis. |
| Step 4 | Plot the points and then draw the bars or lines. |

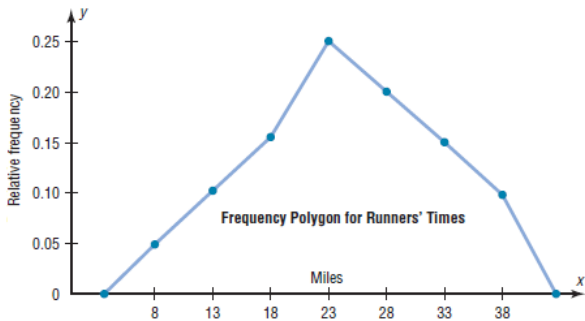
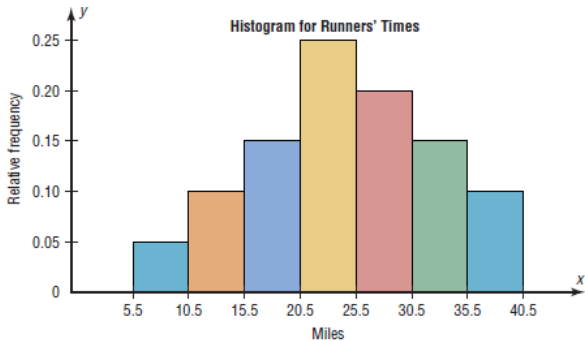
The histogram, the frequency polygon, and the ogive shown previously were constructed by using frequencies in terms of the raw data. These distributions can be converted to distributions using proportions instead of raw data as frequencies. These types of graphs are called relative frequency graphs.

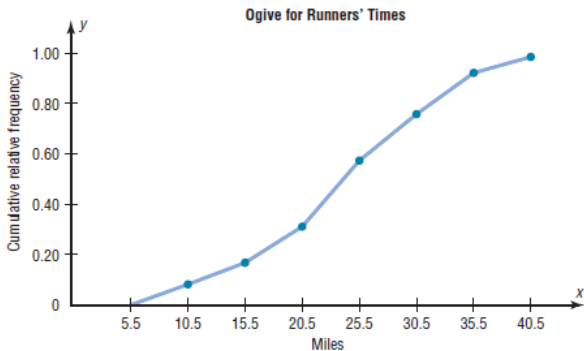
Construct a histogram, frequency polygon, and ogive using relative frequencies for the distribution (shown here) of the miles that 20 randomly selected runners ran during a given week.

Class boundaries	Frequency
5.5–10.5	1
10.5–15.5	2
15.5–20.5	3
20.5–25.5	5
25.5–30.5	4
30.5–35.5	3
35.5–40.5	2
	20

Class boundaries	Midpoints	Relative frequency
5.5–10.5	8	0.05
10.5–15.5	13	0.10
15.5–20.5	18	0.15
20.5–25.5	23	0.25
25.5–30.5	28	0.20
30.5–35.5	33	0.15
35.5–40.5	38	0.10
		<u>1.00</u>

	Cumulative frequency	Cumulative relative frequency
Less than 5.5	0	0.00
Less than 10.5	1	0.05
Less than 15.5	3	0.15
Less than 20.5	6	0.30
Less than 25.5	11	0.55
Less than 30.5	15	0.75
Less than 35.5	18	0.90
Less than 40.5	20	1.00





Distribution Shapes

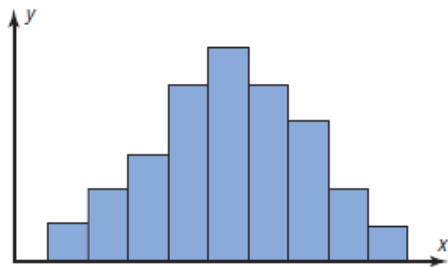
When one is describing data, it is important to be able to recognize the shapes of the distribution values. In later chapters you will see that the shape of a distribution also determines the appropriate statistical methods used to analyze the data.

A distribution can have many shapes, and one method of analyzing a distribution is to draw a histogram or frequency polygon for the distribution. Several of the most common shapes are

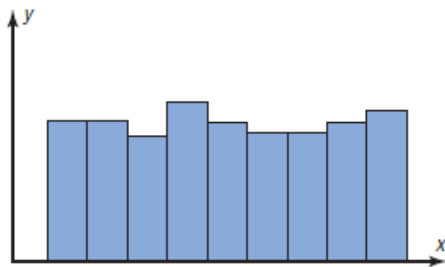
- the bell-shaped or mound-shaped,
- the uniform shaped,
- the J-shaped, the reverse J-shaped,
- the positively or right-skewed shape,
- the negatively or left-skewed shape,
- the bimodal-shaped,
- the U-shaped.

A **bell-shaped distribution** has a single peak and tapers off at either end. It is approximately symmetric; i.e., it is roughly the same on both sides of a line running through the center.

A **uniform distribution** is basically flat or rectangular.



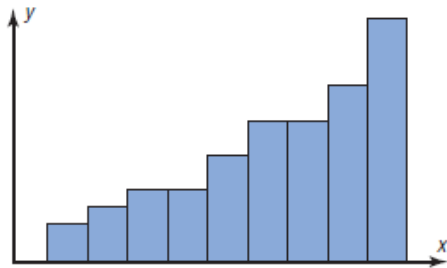
(a) Bell-shaped



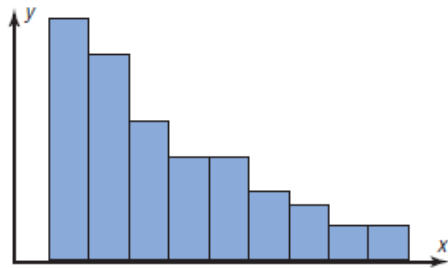
(b) Uniform

A **J-shaped distribution** has a few data values on the left side and increases as one moves to the right.

A **reverse J-shaped distribution** is the opposite of the J-shaped distribution



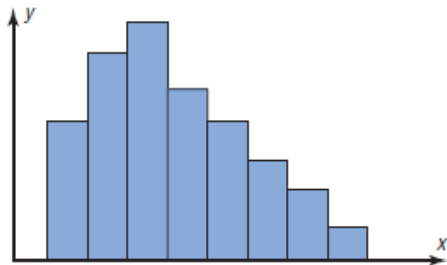
(c) J-shaped



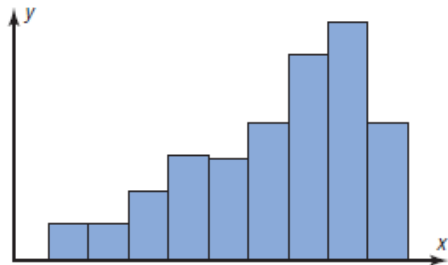
(d) Reverse J-shaped

When the peak of a distribution is to the left and the data values taper off to the right, a distribution is said to be **positively or right-skewed**.

When the data values are clustered to the right and taper off to the left, a distribution is said to be **negatively or left-skewed**.



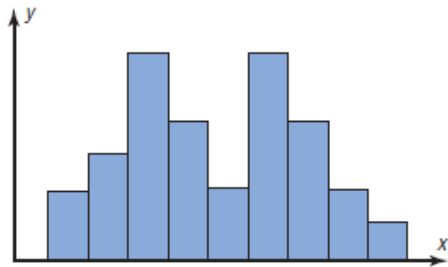
(e) Right-skewed



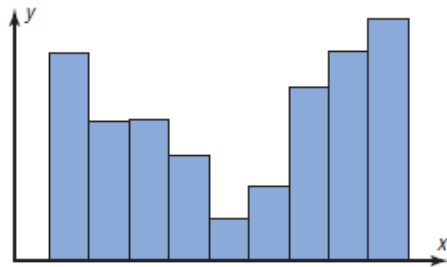
(f) Left-skewed

When a distribution has two peaks of the same height, it is said to be **bimodal**.

Finally, the graph shown is a **U-shaped distribution**.

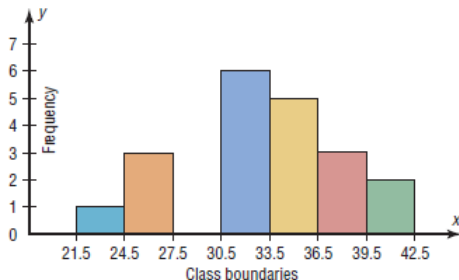


(g) Bimodal



(h) U-shaped

Using the histogram shown here, do the following.



- Construct a frequency distribution; include class limits, class frequencies, midpoints, and cumulative frequencies.
- Construct a frequency polygon.
- Construct an ogive.

Using the results from Exercise, answer these questions.

- How many values are in the class 27.5–30.5?
- How many values fall between 24.5 and 36.5?
- How many values are below 33.5?
- How many values are above 30.5?